



Snowflake Project: Wine Reviews Analysis

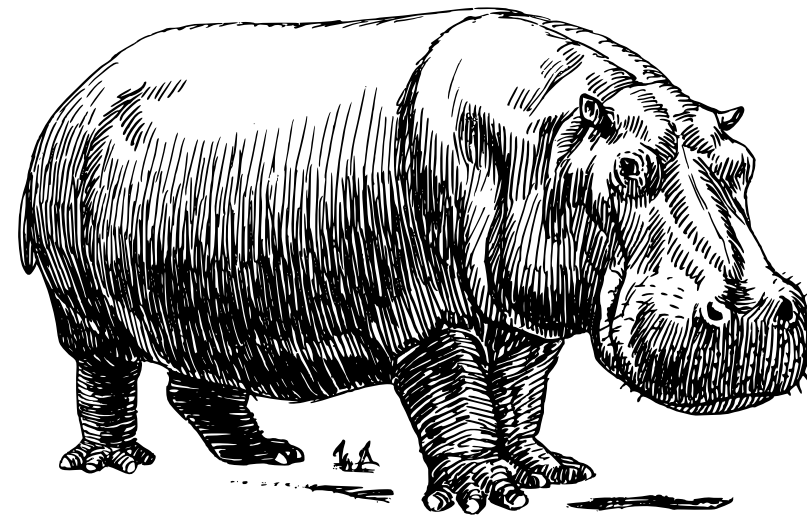
Team: **ROCKSTARS**

Badger



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Medallion Layers (Schemas)

1. **BRONZE:** Raw, immutable data.
2. **SILVER:** Conformed, dimensionalized data.
3. **GOLD:** Aggregated, business-ready data.



-----Schema Creation-----

```
CREATE or replace SCHEMA DB_TEAM_ROCKSTARS.BRONZE;  
CREATE or replace SCHEMA DB_TEAM_ROCKSTARS.SILVER;  
CREATE or replace SCHEMA DB_TEAM_ROCKSTARS.GOLD;
```

Dataset: [Wine Reviews Kaggle](#)



Data Ingestion

1. **Staging:** Data file (winemag-data-130k-v2.csv) is uploaded to CSV_STAGE.
2. **Bronze Load:** All rows are loaded into the WINEMAG_BRONZE_RAW table.

1 file → CSV_STAGE

winemag-data-130k-v2.csv - 50.5MB

×

Select database, schema and stage

Schema DB_TEAM_ROCKSTARS.BRONZE

CSV_STAGE

Specify the path to an existing folder or create a new one (optional)

Example / Path / To / Folder

Once the files finish uploading, your Directory Table will automatically refresh.

Close

Upload

```
687  
688 SELECT * FROM DB_TEAM_ROCKSTARS.BRONZE.WINEMAG_BRONZE_STREAM;  
689  
690
```

Results (just now)

Table Chart

0 rows ⓘ 128ms

	ID	COUNTRY	DESCRIPTION	DESIGNATION	POINTS	PRICE	PROVINCE	REGION_1	REGION_2	TASTER_NAME	TASTER_TWITTER_HANDLE	TITLE
Query produced no results												



Silver Layer Design (Star Schema)

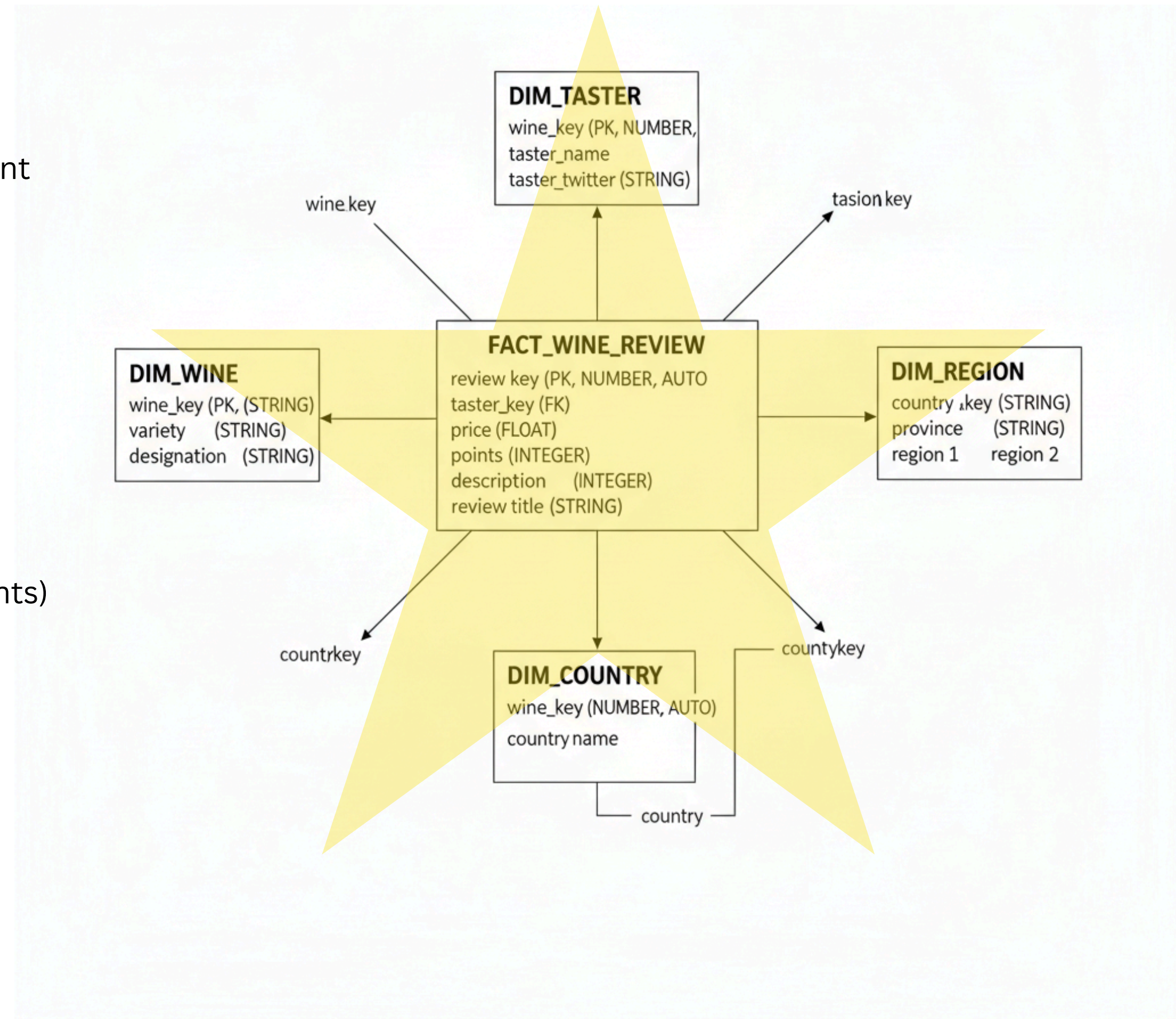
The Silver Layer is designed using a Star Schema for efficient query performance.

- **Dimension Tables (Surrogate Keys):**

- DIM_COUNTRY
- DIM_REGION (Province, Region 1/2)
- DIM_WINE (Variety, Winery, Designation)
- DIM_TASTER (Name, Twitter Handle)

- **Fact Table:**

- FACT_WINE_REVIEW: Stores core metrics (price, points) and Foreign Keys (_key) to all dimension tables.





Silver Layer Population & Consistency

ETL Process: (Bronze to Silver)

Data is cleansed, de-duplicated, and dimensionalized during the move from Bronze to Silver.

1. **Dimension Population Logic (Example: Country):**

Unique country names from the Bronze raw table are inserted, and Snowflake's AUTOINCREMENT assigns the country_key.

2. **Fact Table Population:** The FACT_WINE_REVIEW is populated by joining the Bronze table with newly populated Silver dimension tables to resolve and insert the correct Foreign Keys, ensuring data consistency.

```
//Populate Dim Country
```

```
TRUNCATE TABLE DIM_COUNTRY;
```

```
INSERT INTO DIM_COUNTRY (country_name)
```

```
SELECT DISTINCT
```

```
country
```

```
FROM DB_TEAM_ROCKSTARS.BRONZE.WINEMAG_BRONZE_RAW
```

```
WHERE country IS NOT NULL;
```

```
Select * from DIM_COUNTRY;
```



Gold Layer: Business Use Cases & Metrics

1) TOP_BAND_METRICS

- **Aggregation:** Calculates review_count, avg_points, avg_price, etc., grouped by custom price_band and country.
- **Logic:** Uses a CASE statement on the wine price to categorize reviews into defined price tiers: 'Budget (<15)', 'Mid (15-30)', 'Premium (30-75)', and 'Luxury (>75)'.

2) WINERY_METRICS

- **Aggregation:** Calculates review_count, avg_price, avg_points, etc., grouped by winery, country, and province.
- **Ranking:** Uses the RANK() window function partitioned by country to find the top wineries within each country.

```
CASE
  WHEN f.price IS NULL THEN 'Unknown'
  WHEN f.price < 15 THEN 'Budget (<15)'
  WHEN f.price < 30 THEN 'Mid (15-30)'
  WHEN f.price < 75 THEN 'Premium (30-75)'
  ELSE 'Luxury (>75)'
END AS price_band
FROM DB_TEAM_ROCKSTARS.SILVER.FACT_WINE_REVIEW f
LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_REGION r
  ON f.region_key = r.region_key
LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_COUNTRY c
  ON r.country_key = c.country_key
)
SELECT
  price_band,
  country,

  COUNT(*) AS review_count,
  COUNT(DISTINCT wine_key) AS distinct_wines,
  AVG(points) AS avg_points,
  AVG(price) AS avg_price
FROM fact_enriched
GROUP BY
  price_band,
  country;
```

```
USE SCHEMA GOLD;
--"Which wineries consistently produce the highest-rated wines, and at what price points?"
CREATE OR REPLACE TABLE WINERY_METRICS AS
WITH base AS (
  SELECT
    w.winery,
    c.country_name AS country,
    r.province,

    COUNT(*) AS review_count,
    COUNT(DISTINCT f.wine_key) AS distinct_wines,
    AVG(f.points) AS avg_points,
    AVG(f.price) AS avg_price,
    MIN(f.points) AS min_points,
    MAX(f.points) AS max_points
  FROM DB_TEAM_ROCKSTARS.SILVER.FACT_WINE_REVIEW f
  JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_WINE w
    ON f.wine_key = w.wine_key
  LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_REGION r
    ON f.region_key = r.region_key
  LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_COUNTRY c
    ON r.country_key = c.country_key
  GROUP BY
    w.winery,
    c.country_name,
    r.province
)
SELECT
  *,
  RANK() OVER (
    PARTITION BY country
    ORDER BY avg_points DESC, review_count DESC
  ) AS rank_in_country_by_points
FROM base;

SELECT *
FROM GOLD.WINERY_METRICS
ORDER BY country, rank_in_country_by_points;
```

1. “How do different price tiers perform in terms of quality and volume? Is premium over- or under-represented?”
2. “Which wineries consistently produce the highest-rated wines, and at what price points?”



Gold Layer: Business Use Cases & Metrics

3) TOP_WINES

- **Price Banding:** Custom price categories defined (Budget (<15), Mid (15-30), Premium (30-75), Luxury (>75)).
- **Ranking:** Uses ROW_NUMBER() partitioned by country, variety, and price_band to find the top wines (points ≥90).

```
LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_COUNTRY c
  ON r.country_key = c.country_key
),
ranked AS (
  SELECT
    *,
    ROW_NUMBER() OVER (
      PARTITION BY country, variety, price_band
      ORDER BY points DESC, price ASC
    ) AS rank_within_bucket
  FROM fact_enriched
  WHERE points IS NOT NULL
  AND price IS NOT NULL
  AND points >= 90 -- you can change this cutoff
)
SELECT
  review_key,
  wine_key,
  review_title,
  description,
  winery,
  variety,
  designation,
  country,
  province,
  price,
  points,
  price_band,
  rank_within_bucket
FROM ranked
WHERE rank_within_bucket <= 10; -- top 10 per (country, variety, price_band)
```

```
CREATE OR REPLACE TABLE TOP_WINES AS
WITH fact_enriched AS (
  SELECT
    f.review_key,
    f.wine_key,
    f.points,
    f.price,
    f.review_title,
    f.description,

    w.winery,
    w.variety,
    w.designation,

    c.country_name AS country,
    r.province,

    CASE
      WHEN f.price IS NULL THEN 'Unknown'
      WHEN f.price < 15 THEN 'Budget (<15)'
      WHEN f.price < 30 THEN 'Mid (15-30)'
      WHEN f.price < 75 THEN 'Premium (30-75)'
      ELSE 'Luxury (>75)'
    END AS price_band
  FROM DB_TEAM_ROCKSTARS.SILVER.FACT_WINE_REVIEW f
  JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_WINE w
    ON f.wine_key = w.wine_key
  LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_REGION r
    ON f.region_key = r.region_key
  LEFT JOIN DB_TEAM_ROCKSTARS.SILVER.DIM_COUNTRY c
    ON r.country_key = c.country_key
),
```

3. “Give me a ready-to-use table of ‘Top Wines’ by country / variety / price band for marketing, emails, banners.”



Fully Automated Ingestion Pipeline

The flow ensures new data is seamlessly processed through all three layers.

1) Bronze Ingestion:

Snowpipe Auto-Ingest- A Snowpipe is created with `AUTO_INGEST = TRUE` to automatically copy new files from the external stage into the Bronze raw table.

2) Silver/Gold Propagation:

A Snowflake Stream (`WINEMAG_BRONZE_STREAM`) tracks new INSERTs on the Bronze table.

Stored Procedures (`SP_LOAD_SILVER_TABLES` and `SP_LOAD_GOLD_TABLES`) are manually called or scheduled to consume the stream, loading new data into Silver and recalculating Gold.

Verification: Comparing row counts before and after the stored procedure calls verifies the incremental load (Task 10).

```
-----Create Auto Ingest Pipe-----
```

```
USE SCHEMA BRONZE;
```

```
CREATE OR REPLACE PIPE DB_TEAM_ROCKSTARS.BRONZE.WINEMAG_PIPE  
AUTO_INGEST = TRUE  
AS  
COPY INTO DB_TEAM_ROCKSTARS.BRONZE.WINEMAG_BRONZE_RAW  
FROM @DB_TEAM_ROCKSTARS.BRONZE.CSV_STAGE  
FILE_FORMAT = DB_TEAM_ROCKSTARS.BRONZE.CSV_FORMAT;
```

```
-----Run stored procedures to populate Silver/Gold tables-----
```

```
CALL DB_TEAM_ROCKSTARS.BRONZE.SP_LOAD_SILVER_TABLES();  
CALL DB_TEAM_ROCKSTARS.SILVER.SP_LOAD_GOLD_TABLES();
```



AI SQL - Bronze Layer

Target Column: description (Detailed text of the wine review).

- **AI Functions Applied (Example):**

1. ***SNOWFLAKE.ML.SENTIMENT()***: Derives the emotional tone or sentiment score of the review text.

2. ***SNOWFLAKE.ML.SUMMARIZE()***: Creates a concise summary of the review for quick analysis.

- **Result:** The outputs are stored as additional columns directly on the Bronze table, enriching the raw data for advanced downstream use cases.

The screenshot displays the Snowflake SQL interface. The left sidebar shows the 'Work with data' section with 'Projects' selected. The 'Database Explorer' on the right shows the 'BRONZE' schema. The main editor shows a SQL query that uses `SNOWFLAKE.ML.SUMMARIZE()` and `SNOWFLAKE.ML.SENTIMENT()` functions to process wine review data. The query results are shown in a table with columns: ID, DESCRIPTION, DESCRIPTION_SUMMARY, and DESCRIPTION_SENTIMENT.

```
405 SET
406   description_summary = SNOWFLAKE.ML.SUMMARIZE(description),
407   description_sentiment = SNOWFLAKE.ML.SENTIMENT(description);
408
409 // Check whether columns description_summary and description_sentiment are populated
410 SELECT
411   ID,
412   description,
413   description_summary,
414   description_sentiment
415 FROM WINE_MAG_BRONZE_RAW
416 LIMIT 10;
```

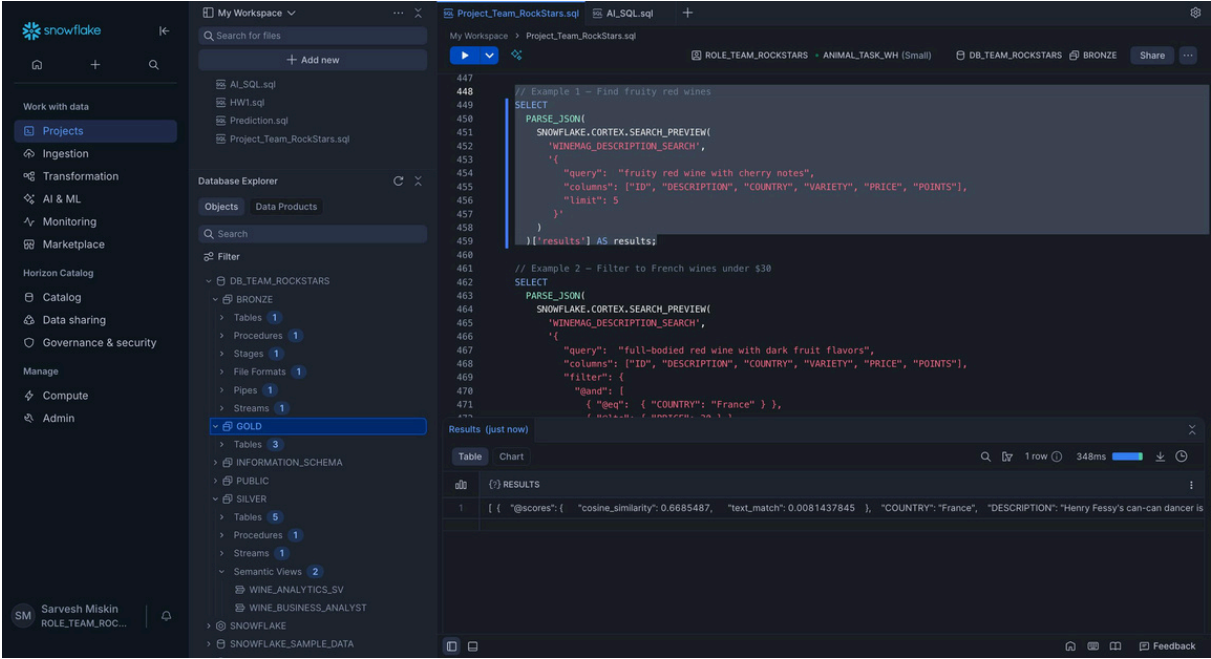
#	ID	DESCRIPTION	DESCRIPTION_SUMMARY	DESCRIPTION_SENTIMENT
1	54169	Aromas of pressed honeysuckle and orchard fruit lead the r	The aroma of honeysuckle and orchard fruit is prominent, w	0.515625
2	54184	Plum, dark berries and a slick streak of almond paste combi	This wine has a combination of plum, dark berries, and alm	0.5625
3	54199	Treated now as a separate wine, rather than a second wine,	Clos du Marquis is now recognized as a distinct wine with it	0.640625
4	54214	Argiano's Suolo is a true standout in the world of Tuscan Sa	Argiano's Suolo is a notable Tuscan Sangiovese with bright	0.8203125
5	54229	Scents and flavors suggest cherry pipe tobacco, along with	The scent and flavor of the tobacco suggest cherry, dried le	-0.40625
6	54244	Charred blackberries and deep caramel notes make up the	This wine has a charred blackberry aroma with deep caram	0.0390625
7	54259	A densely tannic wine, firm at the center and richly fruity on	This wine is dense and tannic with a firm center and fruity e	0.6015625
8	54274	There's a style to this Pinot that takes some understanding	This Pinot Noir has complex flavors of cranberry, pomegran	0.7890625
9	54289	Perfumed wine with some subtle overtones of spice, fresh f	The wine has a perfumed aroma with spice, fruit, and acidit	0.4296875
10	54304	Fresh green apple and riper Cox's Orange Pippin present an	The text describes the aroma and taste of fresh green and r	0.6796875



Snowflake - Cortex Search service

snowflake

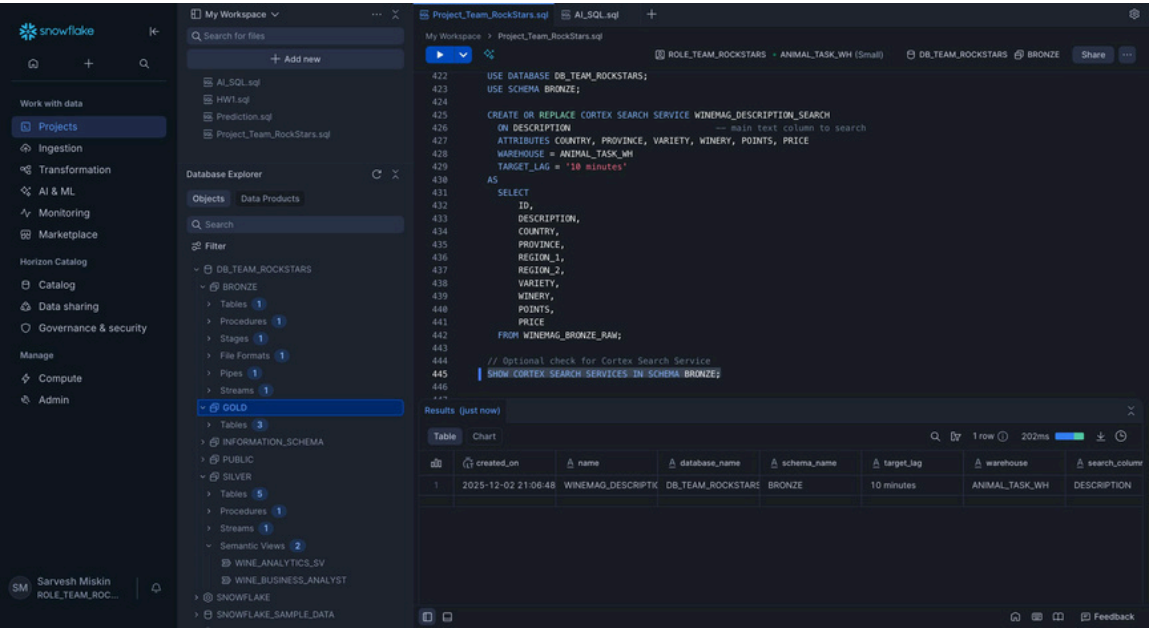
Examples for Search



- **Basic Semantic Search:** The service allows users to find wines based purely on the semantic meaning of descriptive text (e.g., querying for "fruity red wine with cherry notes"), utilizing SNOWFLAKE.CORTEX.SEARCH_PREVIEW to find semantically similar results.

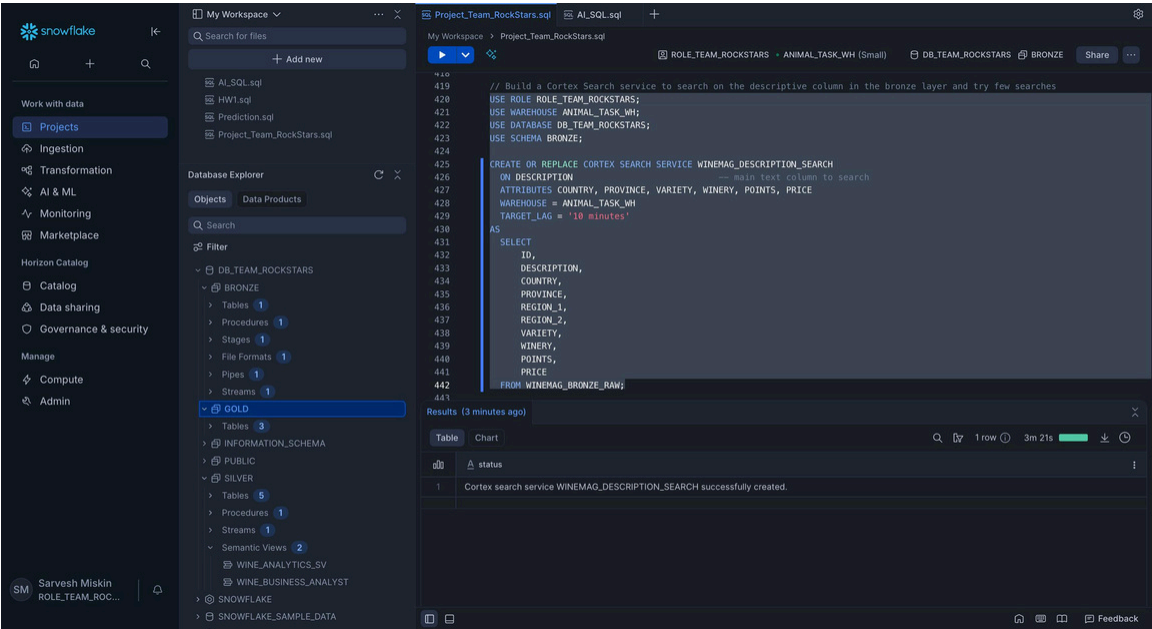
- **Combined Semantic and Structured Filtering:** Cortex Search supports advanced queries that integrate the natural language search (e.g., "full-bodied red wine with dark fruit flavors") with structured filtering via a JSON object, enabling constraints based on dimensional data like COUNTRY='France' and PRICE <= 30.

Cortex Search in Bronze Schema



Cortex Search in Bronze Schema: The service, named WINEMAG_DESCRIPTION_SEARCH, is explicitly created and housed within the Bronze schema (DB_TEAM_ROCKSTARS.BRONZE) and is built upon the description column of the raw table (WINEMAG_BRONZE_RAW).

Building a Cortex Search service



Cortex Search Service: The service is designed to perform searches on the descriptive (unstructured) data in the Bronze layer, allowing users to efficiently find results based on semantic relevance.



Snowflake - Cortex Analyst service

The left screenshot shows the 'Semantic View' editor for 'WINE_ANALYTICS_SV'. It displays a diagram of the semantic model with tables like FACT_TO_WINE and REGION_TO_COUNTRY. The right screenshot shows the 'Playground' view where a SQL query is being executed, and the results of the query are displayed in a table.

- **Semantic View Creation:** The Cortex Analyst service is initialized by creating a Semantic View (WINE_ANALYTICS_SV) on top of the Silver layer tables, explicitly mapping the primary keys of the Fact and Dimension tables (e.g., FACT_WINE_REVIEW, DIM_WINE).
- **Natural Language Querying:** It allows business users to interact with the structured data using natural language questions (e.g., "In France, which wineries have the highest average points, with at least 30 reviews?").
- **Automatic SQL Translation:** The service uses its underlying Large Language Model (LLM) to automatically translate the natural language question into a Semantic Query (SQL), which is then executed against the Silver layer to return structured results.

The screenshot shows the 'Edit Verified Query' view. It displays a natural language question, the generated SQL query, and the results of the query in a table.

WINERY	AVG_POINTS	REVIEW_COUNT
Louis Roederer	93.266667	45
Domaine Weinbach	92.580645	31
Domaine Zind-Humbrecht	92.524752	101
Deutz	92.416667	36
Domaine Christian Moreau	92.264706	34
Domaine Marcel Deiss	92.233333	30
William Fèvre	91.906250	32
Domaine Ostertag	91.846154	52



Snowflake - Streamlit UI service

Work with data

Projects

Ingestion

Transformation

AI & ML

Monitoring

Marketplace

Horizon Catalog

Catalog

Data sharing

Governance & security

Manage

Compute

Admin

SM

Sarvesh Miskin
ROLE_TEAM_ROC...

Welcome to Streamlit Apps!

To let your team create Streamlit Apps, you need to grant the CREATE STREAMLIT privilege! [Here is an example script.](#)

Streamlit Apps

+ Streamlit App

▼

Database All ▼

Owner role All ▼

Group by none ▼

Search

TITLE	LOCATION	VIEWED ↓	UPDATED	OWNER
Wine Insights & Business Metrics Dashboard	DB_TEAM_ROCKSTA...	15 minutes ago	8 minutes ago	ROLE_TEAM_...
SGD	DB_TEAM_DISCOVER...	16 minutes ago	21 hours ago	ROLE_TEAM_...
TV_Shows_Gold_Dashboard	DB_TEAM_QUERYFL...	16 minutes ago	1 hour ago	ROLE_TEAM_...



Streamlit - Winery Performance





Streamlit - Price Tier Reviews





Streamlit - View by Top Wines For Marketing

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Work with data

Projects

Ingestion

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AI & ML

Monitoring

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Horizon Catalog

Catalog

Data sharing

Governance & security

Manage

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<

Streamlit Apps

Wine Insights & Business Metrics Dashboard

Active

Share

Edit

Navigation

Go to

Winery Performance

Price Tiers

Top Wines for Marketing

Country

Argentina

Variety

All

Syrah-Petite Sirah

Syrah-Tempranillo

Syrah-Viognier

Sémillon

Tannat

Tannat-Cabernet

Tannat-Cabernet Franc

Wine Analytics Dashboard – Team Rockstars

Top Wines by Country / Variety / Price Band

Top Wines Table

	REVIEW_KEY	WINE_KEY	REVIEW_TITLE	DESCRIPTION	WINERY	VARIETY	DESIGNATION	
3821	50500	24162	Dolce Stefania 2007 Bonarda (Mendoza)	Chocolaty and malty to start with, then more meaty, rubbery and traditional	Dolce Stefania	Bonarda	None	/
3133	119641	77774	Durigutti 2011 Reserva Bonarda (Mendoza)	The color is deep purple, the nose dense, alluring and hinting at mint, shoe	Durigutti	Bonarda	Reserva	/
3134	88079	82642	La Posta 2014 Armando Bonarda (Mendoza)	Cherry and plum aromas are ripe, sweet and inviting. There's plenty of bod	La Posta	Bonarda	Armando	/
3135	132121	82642	La Posta 2014 Armando Bonarda (Mendoza)	Cherry and plum aromas are ripe, sweet and inviting. There's plenty of bod	La Posta	Bonarda	Armando	/
3136	125841	65658	Zuccardi 2009 Serie A Bonarda (Mendoza)	Deep, compact and rich on the nose, with a big dose of ripe berry aromas.	Zuccardi	Bonarda	Serie A	/
3137	40262	103550	Trapiche 2008 Broquel Bonarda (Mendoza)	Round, thick and pulsing, but not too hard, angular or tannic. The nose brir	Trapiche	Bonarda	Broquel	/
5560	21097	79090	El Enemigo 2012 Bonarda (Mendoza)	Intriguing aromas of lavender, hoja santa and green tobacco blend nicely w	El Enemigo	Bonarda	None	/
5561	19263	79090	El Enemigo 2012 Bonarda (Mendoza)	Intriguing aromas of lavender, hoja santa and green tobacco blend nicely w	El Enemigo	Bonarda	None	/
2460	9005	35391	El Enemigo 2011 Gran Enemigo Red (Mendoza)	Round, earthy blackberry aromas are dressed up in dry spice and oak. A fl	El Enemigo	Bordeaux-style Red Blend	Gran Enemigo	/
2461	132514	111012	Cheval des Andes 2012 Red (Mendoza)	Dark, lightly toasted aromas of licorice root, coconut, blackberry and pepp	Cheval des And	Bordeaux-style Red Blend	None	/

Points vs Price (Filtered Selection)

SM Sarvesh Miskin
ROLE_TEAM_ROC...

Fetching app info...

Thank You



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